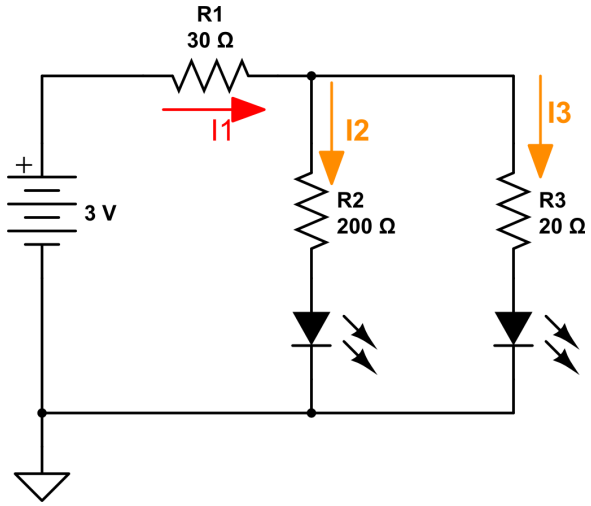
Lab 02 Current and Voltage law (submit before 12:00pm)

Task 1: Wire the circuit on the breadboard (using LED with the same color).



a) Paste your circuit here:

b) Which LED is brighter?

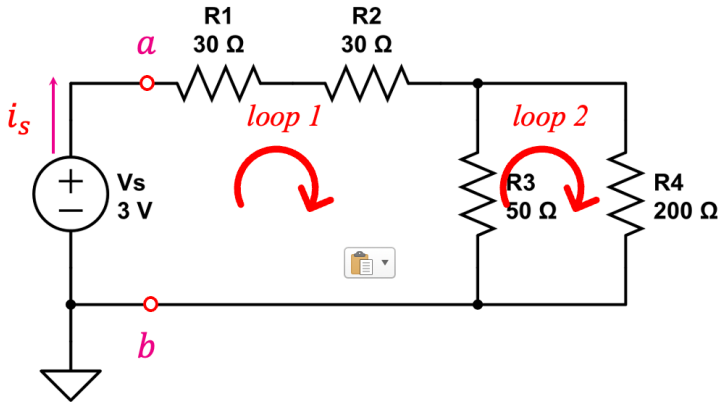
c) Measure I1, I2 and I3 using multimeters, record the readings below:

I1 = _____ *mA,* *I2 =* ____ *mA,* *I3 =* _____*mA*.

d) What is the relationship among I1, I2 and I3?
Do they satisfy Kirchhoff's Current Law (KCL)? (Small discrepancy is acceptable!)

e) Based on the data you collected, explain why one LED is brighter than the other.

Task 2: Wire the circuit on the breadboard, and build it in CircuitLab!



a) Paste pictures of your two circuits here. (one on the breadboard and one in CircuitLab). And record the data in the following table:

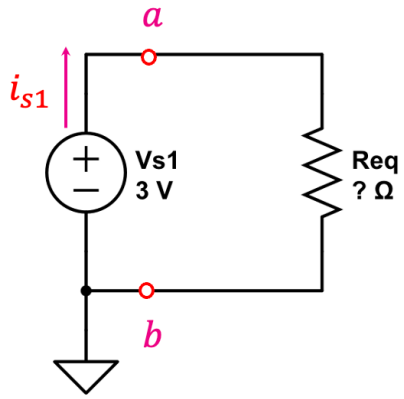
Table 1

	i_s	v_{ab}	v_{R1}	v_{R2}	v_{R3}	v_{R4}
Breadboard						
CircuitLab						

b) Analytically determine the voltages across the resistors (KCL, KVL, Ohm's law), Show all steps clearly.

c) Using the data you recorded, write the KVL equations for Loop 1 and Loop 2. Are the KVL equations satisfied? (Small discrepancy is acceptable!)

d) Calculate the equivalent resistance R_{eq} between terminals *a* and b, as seen from left to right. Show detailed steps.



e) In CircuitLab, replace the resistor network with its equivalent resistor. Record the values of i_{s1} and v_{ab}, and compare them with the results from the original circuit (in Table one). Are the values equivalent? What conclusion can you draw?